ActInf Livestream #008.0: "Scaling active inference" (2019)

Livestream | 1:21:26

all right
hello and welcome everyone to the active
inference live stream
this is active inference live stream
number 8.0
it is november 4th 2020
and i am daniel friedman i'll be doing a
solo contextualizing discussion today
welcome to teamcom everyone we are an
experiment in online team communication
learning and practice related to active
inference you can find us
on our twitter account at
inferenceactive at
activeinference gmail.com at our public
key base team
or at our youtube channel this is a
recorded
and an archived live stream so please
provide us with feedback so that we can
improve on our work
all backgrounds and perspectives are
welcome here and as far as
video etiquette for live streams mute if
there's noise in your background
raise your hand so we can hear from
everyone use respectful speech behavior
etc
so first the announcement is that
we have chosen the papers and the dates
and the time
for the rest of the actin streams for
2020 all meetings
for the rest of 2020 will be from 7 30
to 9 a.m pst

and the papers will be reading number
eight is scaling active inference that's
what this
8.0 is going to be about and that's
going to be on november 10th and 17th
paper 9 will be the projective
consciousness model and phenomenal
selfhood a 2018 paper
paper 10 is going to be a variational
approach to scripts
paper 11 is sophisticated active
inference effective information sorry
simulating anticipatory effective
dynamics of imagining future events
and you can see the dates that all these
events are so
set aside at times if it's possible for
you to participate
and if you have a time zone or kind of
event that you want to do that's not
reflected here just let us know
and there's our twitter address all
right
so here's what's going to happen in
active stream 8.0 this one
the goal of this talk is going to be to
set the context for 8.1 and 8.2 which
are going to be on this
same paper scaling active inference this
is a paper by
alex chance baltierri seth and buckley
from 2019
with the archive 1911.10601
the video is an introduction to the
context of some of these ideas it's not
a review
or a final word definitely i learned a
lot just reading through the paper and

researching for this presentation
so the idea is that this video will
contextualize some of the ideas
math and notation and vocabulary of the
shantz paper
and the video should be accessible
though this is also hopefully cool
and cutting edge research
and the punch line and don't worry if it
doesn't make sense yet or the
implications aren't clear yet
is that active inference is scalable and
it's homologous too
so it's similar to and potentially
preferable to other
common algorithms in a similar space
like control theory or machine learning
so in 8.0 the first section
is going to be some background on all
the keywords that they used
in this paper that they provided and
then talk about the goals the abstract
and the roadmap
then the second part of 8.0 will be the
key equations annotation
and quotations walkthrough and we're
going to do like the 80
20. so most of the notation most of the
meaning but
not all the sections not all the symbols
and then talk about figure 1 and figure
2
and what they represent and how that
supports the conclusions of the paper
and then in 8.1 and 8.2 we will all come
together to discuss the same paper
so save and submit your questions or put
them as a comment and then get in touch

if you want to participate in this one
okay so let's start with the keywords
so these were just the keywords that the
paper provided so i would think of them
these are the topics that
this research is going to be on the
cutting edge with respect to that field
hopefully so they are artificial
intelligence
machine learning and they mention also
reinforcement learning
and model based reinforcement learning
and then
systems and control and information
theory and then a keyword that wasn't on
the
paper but we can just add here is of
course active inference and the free
energy principle
so um each of these keywords definitely
you could have a
course or a phd on so
each one of them is going to get about
one slide so
of course go deeper into the resources
mentioned
or find other courses that teach these
techniques because there's a lot to them
i'm just going to kind of
go through them in a way that builds
towards where we're going
with the math and with this research
paper and just know that there's other
people's perspectives on these topics
too
so first artificial intelligence
wikipedia when we last checked it
was saying that artificial intelligence

or ai
is intelligence that is demonstrated by
machines unlike the
natural intelligence displayed by humans
and animals
so even using this definition or perhaps
an alternate one which i'll get to in a
second
we can say that what ai results in today
2020
is it results in maps policies
individual and collective policies
recommendation engines it results in
encryption techniques and security
it relates to classification algorithms
and things that influence us
us all now where i would come at this
is by saying well ai is made and used by
humans as part of their
natural world and everything is natural
in that it exists in the world and so by
positing that there's an artificial type
of cognition
that is somehow uh distinct rather than
extended or embedded
enacted by humans does a few things
that aren't very helpful and there's
been some excellent
other discussions this topic just to
move it in a positive direction
i would just say that often other
phrasing can be more helpful
than calling it ai so here's one example
what we're going to talk about a lot
today is computer statistics
so using computers to do statistics but
basically things that pen and paper
could do slowly so things that are about

matrix calculations are about
statistics with computers on big data
another way that ai can be meant today
is like
human in the loop ai so human in the
loop ai could describe
a map or a recommendation engine to
emphasize that it's human agency all the
way through
in the use affordances as well as
hopefully in what the designers
considered
another phrasing of ai that can be
helpful
and also center the role of the human in
deciding
how to use it and how to design it is
intelligence augmentation which is
clarifying that as us who are
augmented or even a more general
phrasing than ai would just be the human
technological niche
which would also include physical
aspects of our niche for example what we
talked about in 7.2
but that would include effects that
aren't um just
simply in the sort of silicon chips and
their interactions that ai
captures all right machine learning
so machine learning there's a lot of
things to say about it it's a big area
but the main
machine learning topics that come into
play in this paper
are reinforcement learning and model
based reinforcement learning
so the simple reinforcement learner

is an agent in an environment and the
agent
takes actions a that act on the
environment
and then the environment sends back
states and rewards
and so sometimes it sends back just
states directly
and then the reward is um understood
through a layer
of perception by the agent other times
the environment directly sends reward
back
where model based reinforcement learning
builds on that is by the introduction
of this model which is outlined in red
here
and in these types of model based
reinforcement learning
rather than learning these the simple
raw connection
between successful behavior and
reward and learning that as a behavioral
correlation
the model based reinforcement learner is
able to have a model of
oh for example running is painful now
but then i'll feel better
later and so then it allows one to
pursue goals that are transiently
uh unpleasant in order to get at
larger term goals so think about in a
little maze you know trying to get
not just greedily towards the exit but
trying to take a step back
to take two steps forward so the
similarities are that agents are acting
on or within the environment and that

their actions arise from policies which
are models of action
so what is and isn't likely and those
can be
um concurrent with what the organism
actually can or can't do
quote but at the very least it's what
ends up getting implemented by this
organism the agents modify their action
policies based upon
their reward from the environment either
in a direct way or in a symbolic way
or in a model based way they learn
or to use the bayesian phrasing they
update their model
so in any case it's going to be
computational because we're going to be
using computers and
math to describe it but then bayesians
would say like
the priors updated with evidence to
generate the posterior
and there's various ways that this
learning or update can be done
another similarity is that the
environment sends rewards or signals to
the agent
and one area of debate or question to
think about
is is the observation the reward or how
is it the reward
is the learning itself the reward
there's machine learning algorithms that
have that kind of approach as well
and then does the observation symbolize
or signal
the reward or how do we think about that
those are kind of the ways that

reinforcement learning and behavioral
algorithms
can be applied to both abstract data
classification
approaches clustering and things like
that as well as direct plans for action
and it turns out that active inference
is going to relate to reinforcement
learning an interesting way that's why
the authors made it a keyword
all right now in a modern context this
reinforcement learning
is going to be implemented with a
for example neural network that in this
case we have the environment
sending the state to state s on the
agent and that's going to go through
some sort of deep neural network dnn and
result in some policy selection
so neural networks are inside the agent
that's a way that we can implement it
with basically software that's available
and that allows for some types of
scalable in the sense that we know how
there's uh going to be a relationship
between how many models our parameters
have how many how big our model is and
how effective it's going to be and then
it also allows for non-linear
inference not just a simple linear
regressor let's say
and sometimes the little and sometimes
the big theta is used for parameters but
we'll get to that later
and the neural network can be model free
so it can just be
sort of let loose to find the patterns
in the data

or it can be using a underlying model
so if a certain type of event is a
priority can be much more likely
those kinds of observations can be used
to in a more weighted way
one example of another
modern reinforcement learning is this q
learning which was used for some games
and some types of things like that
recently
and the insight is just another way of
doing it
is by mapping the cue table
which is the mapping of state action and
reward
so that would be mapping your current
state and your behavior and that say
like oh i'm tired but i'm going to keep
running because it's rewarding
let's say but in some mathematical way
and then the deep q learning also
represented here is
when again it's not just a table but
it's a neural network so kind of the
modern inside just replace any other
statistical module with a deep learning
something
and uh that allows for apparently some
more nuanced policies to arise
and then just one slide just to throw up
there you can pause if you want to look
at it but it's from a paper called deep
reinforcement learning
and it shows just the depth of how many
different areas this is kind of related
to
so these are the kind of structure of
the problems and the areas that

this paper is getting at with these
keywords
okay so let's start from this model
based reinforcement learning framework
and then talk about systems and control
so in this framing we can look at the
model based reinforcement learning
and the green green and the blue model
free reinforcement learning so model
free again
is the direct mapping from experience
onto policy
and the increasingly abstract model
based reinforcement learning is based
upon the supervised learning of
experience
into updates for the transition model
generative model of the world
and then using that to implement some
sort of planning process
resulting in some policy selection
so control systems theory is about
planning for action amidst uncertainty
because you have to plan for action and
there's going to be uncertainty
and there's various kinds of uncertainty
sometimes they're metaphors for each
other other times the methods are
transferable
other times one system will suffer from
one type of these challenges
versus another but it includes systems
that are chaotic like a double pendulum
stochastic multi-scale partially
observed noisy
etc so if you can't act you can't
control you can only watch and that's
really what differentiates

control theory from just statistics
because statistics is descriptive
but systems and control approaches are
more oriented towards
understanding the dynamics of the
systems in a way that can be intervened
in
so modeling this planning process and
the policy explicitly
and then also of course the caveat that
not present in many of these models is
the strategic
axiom that non-action can be a form of
action
so the summary here is that we want to
have a mathematical
model a way to talk about how
a system has to supervise or model their
self their action affordances and their
system and their world well
that's the good regulator and also have
this
action policy plan so supervised
learning
is about supervising your input and
appropriately updating your generative
model of the world
and then what is your generative model
the world for it's gonna be about this
policy selection and here's a more
classic control theory diagram with the
way that the
system measuring element controller and
effector are related
all right information theory
another keyword so there's probably
another
500 hours of youtube to watch on info

theory and a lot you could say about a
lot of different areas
um a simple way to phrase it is that
information
is the reduction of uncertainty whether
it's actionable or not
so it doesn't have to be connected to
control theory it can just be purely
descriptive like taking the
shannon entropy of some dna strand
information is always contextual it's
relational it's model dependent there's
a lot of critiques of this
sort of naive shannon interpretation
of information theory sometimes and the
measurement of or the comparison of
information is challenging like if you
calculate the entropy of the
biodiversity of insects
and plants on the same island they're
not necessarily on the same scale et
cetera et cetera
and uh there's a lot of areas areas that
info theory touches on like semiotics
and semantics measure theory
dynamical systems and signal processing
reduction of uncertainty is also not
truth because of course
wrong and precise estimates do exist and
precise estimates of external states
don't really facilitate
effective action in many cases a lot of
other related areas are the
quantification
storage and communication of information
the measurement comparison mentioned
before how do you store
and transmit information through time

that's like memory
how do you communicate that's about
transmitting information through time
and space
protocols for communication
and when information gleaned by a system
reduces its uncertainty about world
states or causes or relationships
it can facilitate the updating of
effective policy models and therefore
action
in some cases but not all cases and when
the action is effective
it can be rewarding or sustainable or
high fitness there's various ways that
you can think about that kind of a
success depending on the mechanism
scaled system
and then the parameters are updated by
this rewarding
so whether it's which genotypes of the
plant
are successful in a certain niche or
whether it's which computer virus is
mutating the best
it's this sort of learning slash
development evolution process that
ultimately we want to kind of move
towards an integration of
all right just to talk a little bit more
about information theory and also
introduced this conditional notation
just for those who may or may not have
seen it before
why are we talking about information
theory here because
again when we reduce our uncertainty
about

causal relationships and the statistical
regularities in the world
we can enact better policies so let's
get a grip on
some basic information theory concepts
but also introduce this notion of a
conditional
so the notation a vertical line b
means a given b so the part after the
vertical line
is the part you're conditioning on h of
a
is the information content how
surprising
is the random variable a in units or in
bits
and just to kind of combine these two
bullet points
h of a given b is the information
content of
a conditioned on b sorry there's a typo
on there
information of of a semicolon
b is the mutual information between a
and b so that's
shown in these sort of single overlap
spaces on the venn diagram below
and then also there's a triple overlap
and you can also have overlap and
conditionals
so the way that we're going to think
about mathematics is just sort of like
nested
operations that hold different kinds of
transformations and potential
and there's a lot of cool areas of
information theory
james gleek's a great writer and

this book by stone in the middle is a
good introduction and then this book
buys a neil at all
algorithmic information dynamics is also
a cool
application and theory so let's return
to a little variant of a slide that
we've looked at kind of before
which is this inactive versus bayesian
dichotomy divide that ultimately active
is going to try to integrate
so in the inactive worldview we have the
agent in the world and they're enacting
in this exit nexus of action and
perception
and the outcome of this interaction is
the embodied and ecological action
sequence or policy from an embedded
agent who
makes slash is their behavior and
just to go one more level now from its
control theory perspective
into the agent we can see that the agent
has the sensory states on the in
input side then they have their causal
model of the world
and then their policy model in some
sense these could be like
one integrated unit or not we'll get
there later
and then we can line that up with the
bayesian
structural representationalist view so
in this
view you have the data the observation
which are sort of the base level
observations or parameters in the model
and then it through a recognition model

hyper parameters are updated which
result in
a generative model of the data and this
kind of
a expectation maximization scheme
results in the statistical convergence
of a multi-level model
that's representing hence the
representationalist
structures of the world hence the
structural part
maybe and then just to line it up we can
think about that model that's a bayesian
control agent let's say
it's going to have sensory states what
is my
robot sensor detecting what is the
causal model for what that means for the
pose of the robot and then the policy
selection how are we going to
correct so these uh both can be
these two sides of the dashed line can
kind of be
internally consistent and we want to
return to this question which is about
how
the free energy principle and active
inference is going to say something
about
the relationship between the world and
the agents
so we have action and perception and we
want them to be symmetrical in this kind
of
niche concept way we've talked about we
want to have
the free energy principle be an
axiomatic

set we want to build on top with a
multi-scale active inference
perspective this is sort of where we
want to be moving towards how has active
inference
stepped into this gap and built up
towards this goal
so active inference 1.0 and
open to people building on slash
correcting me on any of these parts here
this is just very
rough epoch level active inference 1.0
is a implementation of a machine
learning type algorithm that builds on
free energy principle as well as the
inactivist approach and it relates the
attunement of local sensory states and
policies
so in that way it's kind of like the
model free
reinforcement learner and it focused on
very discrete cases
so just like on a grid for example
and it used pretty simple expectation
maximization algorithms and
had relatively short range spatial and
or temporal
reach or depth it was um more of a
uh let me see if my
video not sure what happened here
um
sorry about that
so i'm gonna just
re-add my camera
sorry
all right well c'est la vie
anyways active 2.0 added
temporal depth and niche interactions

into the equations with a longer term
planning phase and also added in this
aspect of agent embeddedness and
introduced the idea of valence and
learning into
the formulation and this is kind of
where active inference 3.0
and beyond go so one direction they've
been going
is towards learning and affect and the
sophisticated or counterfactual aspects
of
action here's a citation
and then another direction that it was
going towards is
reflected by this scaling active
inference paper
which is going to introduce the idea of
the high dimensional data
the continuous variables explore exploit
behavior and
also homology to reinforcement learning
and working at scale
so for me it was like the hero's journey
reading this paper because this is just
a framework for a completion
story so this is the scaling active
inference
journey so it starts off with a call to
adventure
which is the question how can we model
complex control tasks with active
inference
here's the beginning of the
transformation the challenge is in the
temptation
should should one go to sleep or should
they read

fristen's bibliography
the helper comes into the picture here's
alec
we have a transformative moment when we
read and when we understand
the paper scaling active inference then
we have our
movement from dark to light from
reinforcement learning to active
inference
from failure to control along the way we
receive the gift of meeting new friends
discussing cool ideas
having impact on systems and making
first and memes
of course so let's get to the paper
so scaling active inference is the paper
the goal of the paper
is presented as we present a model
of active inference that is applicable
in high dimensional control tasks with
both continuous states and actions
our model builds upon previous attempts
to scale active inference by including
an efficient planning algorithm as well
as the quantification and active
resolution of model uncertainty
our model makes two primary
contributions first we showed
that the full active inference construct
can be scaled to the kinds of tasks
considered in the rl literature
this involved extending previous models
of deep active inference to include
model uncertainty and expected info gain
second we highlight the overlap between
active inference state-of-the-art
approach

to model-based reinforcement learning so
the non-active inference phrasing here
is how can we apply active inference to
challenging control tasks
and connect it formally to reinforcement
learning so the abstract was
in reinforcement learning agents often
operate in partially observed and
uncertain environments
model based reinforcement learning
suggests that this is best achieved
by learning and exploiting a
probabilistic model of the world
active inference is an emerging
normative framework in cognitive and
computational neuroscience
that offers a unifying account of how
biological agents achieve this
on this framework inference learning and
action
emerge from a single imperative to
maximize the bayesian evidence for a
shared niche of the world
however implementations of this process
have thus far been restricted to low
dimensional and idealized situations
here we present a working implementation
of active inference that applies to high
dimensional tasks with proof of
principle results
demonstrating efficient exploration and
an order of magnitude
decrease in sample efficiency over
strong model-free baselines
our results demonstrate the feasibility
of applying active inference at scale
and highlight the operational homologies
between active inference

and current model based approaches to
reinforcement learning
okay so for some of you
that might be a lot of new ideas others
may recognize a lot in there so
let's just lay out the road map for how
we're gonna get from
a to z and then hopefully
it will be approachable from all these
different
angles section one
is an introduction and two is about
active inference
and what work has been done in the area
three
is the model which is what we're going
to focus on in this
discussion and 3.1 is the generative
model and the recognition
distribution those are the p and q
models
then there's a section on learning and
inference policy selection
trajectory sampling and expected free
energy
as well as a section on the fully
observed model those are the sections we
won't go
that much into today then there are the
experiments
related to exploration and exploitation
and the two figures related to the
comparison of exploration strategies and
the comparison of performance onto
continuous control tasks
then the relationship with previous work
specifically deep
active inference model based

reinforcement learning and information
gain
theory then there's a discussion and
conclusion
all right
this quote which you can pause
to read in full i'm just going to pick
up on the
end of the top paragraph while this
approach the previous work provides an
elegant framework for evaluating
expected free energy it can only be
applied in discrete state and action
spaces
meaning it is not directly applicable to
the high dimension states and continuous
actions considered in reinforcement
learning benchmarks
so this is one reason why active
inference discrete
states based learners have not been
compared
in the author's phrasing is that
the reinforcement learning benchmarks
that are used to test
the adequateness or the efficacy of
different reinforcement learning
control theory algorithms are
are continuous tasks like the kind that
we'll talk about in the figures
and so what this paper is going to do is
employ an approach called
amortized inference which we'll talk
about more in a little bit
which utilizes functional approximators
i.e
neural networks to parameterize
distributions

free energy is then minimized with
respect to the parameters of the
function approximators not the
variational parameters themselves
and then they restate what each of the
sections and three are
here is a notation reference sheet it's
not every single
parameter especially the statistical
parameters that are distributional and
a lot of the later sections too but this
is
the sort of parameters that we're going
to see a bunch
in the coming slides and
we can really
just already look at what a lot of the
topics that the model is going to cover
are so for example the policy of the
agent
the states that it's trying to estimate
and the true states with the hats it's
going to be through time it's going to
involve observations
we're kind of seeing a lot of the
ingredients of the reinforcement
learning algorithms
that we were talking about earlier so
let's
talk about this amortized inference
here's what they write the author's
chance at all
amortizing the inference procedure
offers several benefits
for ex for instance the number of
parameters remains constant with respect
to the size
of the data and inference can be

achieved through a single forward pass
of the network rather than let's say
back propagation training
moreover while the amount of information
encoded about variables is fixed
the conditional relationship between
variables can be arbitrarily complex
in equation three which we'll get to the
parameters of the transition
distribution data are themselves
random variables in the current context
these parameters are the weights of a
neural network f
this approach allows the uncertainty
about these parameters to be quantified
and cast learning as a process of
variational inference
the prior probability of theta is given
by a standard gaussian which acts like a
regularizer during learning
um and this just talks about how free
energy is going to be used to select the
action
policy so what is this amortized
inference why is it being used
these papers on the bottom have a lot
more information but in amortized
inference the number of parameters used
is flexible you can have a small or a
large model
but what's good about it is that you can
also
set the size of the model to not change
even as more data is input so this
allows you to train on devices that you
know are going to
have limited computational power or use
a very large but

defined model size to implement on
larger scale networks let's say
again the model can be trained with a
single forward pass rather than more
complex games
and then also there's all this degree of
freedom when you can design the neural
network because there's a lot of
approaches that can be implemented with
neural networks
the general framing that they're going
to be thinking about for this whole
mathematical journey that we're about to
describe
is a partially observed markov decision
process so these are the key variables
that are going to be defining this
organism-centric approach let's say
at each time step t so that one's easy
to remember the true state of the
environment which is
s with a hat hat means that it's the
real state and
the states are what are the important
things those are
like is the bus gonna hit me or not
gonna hit me let's say
but they could be various other things
so we're keeping it general
and then that s hat sub t so the real
states through time
exists in a state space with a certain
dimensionality
and there's also a real transition
dynamics of the real states that's
related to
um this p s with a hat t given
the previous times and the action so

that's saying that
the time evolution of the system the
real one with a hat s of t
is related let me see if i can actually
get the laser
yep the s of t with a hat
is related to the current time
and the previous time and the actions
and that has also a dimensionality
and agents do not have access to the
true state of the environment but
might instead receive observations so oh
observations
it's a good easy one and those have a
certain dimensionality which are
generated to an actual
observation distribution t which is
related to
how observations are conditioned on real
states of the world so if it's really
sunny outside
then you're really going to get photons
if you look at the sun
as such um again this is because agents
only get observations and not the
absolute truth from the world
agents must operate on their beliefs
about states so that's
s without a hat through time those have
a certain dimensionality
about the true environmental states as
with a hat
and then there's a difference between
the true dynamics without
italics and the model of the dynamics
which are always with the italics
and here's how those equations are
linked here's where active inference

comes into play
active inference proposes that agents
implement an update a generative model
of their world so italics
p that means the model of the world of
observations
states policies and parameters of the
world
where the tilde is the sequence of
variables through time
so those are the observations and states
through time and pi is the policy
like if i have the policy of running
every day then my observations through
time are going to look like this
you can imagine that one and theta
um with italics is belonging to a larger
version of theta capital theta and that
denotes parameters of the generative
model which are themselves random
variables
and also agents maintain a recognition
distribution this is like p and q
so mind the p's and q's of the states
through time
and the policies and the theta
parameters of the world representing the
agent's
beliefs over their states policies and
model parameters
all right so in this
part we're going to look at the p model
remember it's the ps and the qs
so the p is a agent implementation
that's why it's in italics
of a model of the observations through
time
o with the tilde states through time s

with a tilde
policy and model parameters so that's
what
this model is about
and it's implemented by the agent that
we care about
here's what each of these lines says so
the first line
on the left side of the equation says p
which is
that function that we have in big on the
top
is equivalent to the
probability model of the
parameters theta take those out
then the probability of the policy is so
if it's not possible for me to jump
a thousand feet then it's not going to
be part of my probability distribution
if it is part of my power to jump a
thousand feet then i'll consider it in
my
policy estimations and then the big
pi it's kind of like how sigma adds
stuff
up pi multiplies stuff up and so this is
saying
multiply it all out through time
starting from the beginning t
equals one and look at the p
the agent model of o
through time observations through time
given states through time so the first
part is just
what's the probability of given that i'm
um healthy that i'm seeing this
state and then that's multiplied
at each time point by the

ways in which states s t are
conditioned on the prior states and the
prior policies
and the parameters of the model so how
all those things
are linked multiplied out through time
is
what at a first pass describes this big
equation on the top the total model of
os pi
theta then
to dive into this p of observations
given states
through time model that the agent
implements n
big n notation means normal and so this
is saying
the model of observations given the
state estimates
is a normal distribution again
with observations through time with a
certain
mean and variance mu and sigma with a
certain parameter this lambda and
together
these two this kind of like dupl
tuple of variables
constitute this function lambda
so the way that we're going to estimate
these mean
variants may or may not be a least
square it might be
not not an l2 norm we could
use a functional approach here that's
the insight of amortization
the probability of states that
probability
transition model of how the agent

believes states through time
to be linked to conditional two um
states at prior time and policy at prior
time and then the parameters of the
world
is related to another normal
distribution
of states through time with another mu
and sigma with a different
subscript and for
those mu and sigma we have f um
with the sub theta of how the uh
states and the policies are related
so this is another just sort of way to
estimate
the variance of different kinds of
events
happening therefore weight sensory
inputs
together the state estimations
distribution
are again yeah estimated by a function
p of theta is given
as a normal distribution with a mean of
zero and this
identity matrix of covariance i believe
and uh i think this is just reflecting
a neutral initiating approach or
regularization but
i i'm this is not 100 sure about that
and this is reflecting that the pr the
probability model of
policies is related to this
max function so it's a sigma but it's
not the same
as sigma squared in a statistical model
and it's the soft max function of the
negative

expected free energy of the policy
which we'll talk about in a later
section
so that was our p let's look at q
q is this agent recognition distribution
the q with the italic
of states through time policy and model
parameters
here's how that one is defined
so q is the probability of the
parameters theta
multiplied by the probability of the
policies pi and then that same
multiplication through time for all time
of q which is the model of states given
observations
so previously
we had observed p-o-s the second line
here
probability of observations given states
now we have um q s o
so this is kind of going the other way
two of theta
is a normal distribution kind of as we
saw before
with its mean and variance parameters
same as the policy
and same as the uh the q
model of the states given the
observations
and it turns out that those mu and sigma
are gonna also have a functional that's
going to approximate them
so to make active inference applicable
to the kinds of tasks that are
considered in reinforcement learning
the authors treated reward signals
observations

as observations in a separate modality
so that extends the generative model to
include
a additional scalar gaussian so that's
another
reward value of scalars like a single
number over reward observations with a
unit variance in mean
and that allows them to wrap this neural
network
which is uh f sub alpha of the states
which is like this reward
fully connected neural network with the
parameters that they're then going to
train
so that's really where the formal
homology arises from the model based
reinforcement learning
is that now we have pretty much the same
architecture that was specified earlier
in this discussion where the agent state
is then of sensory input is run through
this dnn
and that results in this output of the
policy
or the reward but it's a little bit
different but that's
how we see a lot of the same homologies
all right so what do the
learners do as new observations are
sampled
agents update the parameters of the
recognition distribution
to minimize variational free energy or f
so the f of the observation is
the expectation um okay well
let me finish reading first this makes
the

what this f is going to do is make the
recognition distribution
q converge towards an approximation of
the intractable posterior distribution
p thereby implementing a tractable form
of approximate bayesian inference
alright so f of
o through time with a tilde is the
variational
free energy of the observations through
time
so what is this magical free energy
not magical it's just this equation in
this case
it is an expectation that is conditioned
on a specific recognition model
which is this expectation of q that's
going to be
related to this um s
pi policy and theta parameters
uh variables what is this expectation
gonna be of
that's everything that's about to be in
square brackets
this expectation is going to be the
natural log ln
which is a way to um turn a
question about maximizing the goodness
of a model's fit
into this minimizing the negative
sometimes there's a few
ways that the natural law can help the
expectation
is going to be the natural log of the
recognition model
minus the natural log of the generative
model
so again that's the part that we cue

that we can learn on
and then the part p that is
characterized as intractable
it turns out that this expectation is
always bigger than or equal to the
natural log of p of the observations
so that's the just the p model of
observations likelihood
by minimizing the free energy of the
observations through time
the agent converges heuristically
towards this intractable p
distribution given that
q is only over the state's
policies and parameters and that p is
really including and focusing on the
observations
what this allows for the agent to do is
to focus on modeling
the states and the policies and the
parameters that are relevant via queue
and then potentially use math slash
inactive behavior in this model to
attractively include condition upon or
learn from its observations
so we could bring previous knowledge to
the table just like bayesian models
allow for
but also have a way to deal with sparse
or dense information
as it is
okay crucially active inference also
proposes that an agent's goals and
desires are encoded in the generative
model as prior preferences for favorable
observations
i.e blood temperature at 37 celsius
free energy then provides a proxy for

how surprising ie unlikely some
observations are under the agent's model
while minimizing equation 1 provides an
estimate for how surprising some of our
observations are it cannot reduce the
quantity directly
so here equation one we defined what f
was but it's just a definition and a
bounding
of what f is it cannot reduce this
quantity directly so it's not really
related to
policy as directly approaches
to achieve this agents must change their
observations through action
acting to minimize variational free
energy ensures the minimization of
surprise
surprisal negative natural log of the
probability model of observations
or the maximization of the bayesian
model evidence
p of observations through time since
free energy
provides an upper bound on surprisal
that was equation one
active inference therefore proposes that
agents select policy
in order to minimize expected free
energy
fancy g where the expected free energy
for a given policy pi at some future
time is
going to be defined this way so
we want to do more than just update
our internal model to bound our surprise
on our observations we don't just want
to fit the statistical descriptor model

of the world
we want to reduce the uncertainty in the
future which is a future time point t
tau through the policy pi that we take
now
this minimizable expected free energy
function fancy g
is going to take in the arguments of the
policy selection
and the future time step so fancy g
fancy pi
fancy t that's the free energy function
in this case
and what that value
which optimization on it is going to
allow for effective action
is going to be defined as is the
expectation
that's going to be related to q model
which is about this observation states
and parameters given policy models so
given that my policy is this
then how will the states and transitions
of the world
appear and it's going to be the
expectation of this quantity that's in
brackets
it's going to be the log of states and
parameters
so that's the q of the s sub t
and parameters given the policy
so that's what states will i be in and
what will i think about
um along the way potentially these are
just
first pass verbal explanations just look
at the math but
what states and parameters am i in now

[Music]
at a future time given my policy so if i
go out on a walk every day how will it
influence my
states what does my model have to be at
that time
and then that quantity minus natural log
of p
the observations states in the model
given policy so that's the part that we
want is actually the observations to be
the correct values but what we can do
is um just condition the states on
policy
and that g value is always greater than
the
negative that's this last part of
equation two then the negative
expectation
of this q which is the observations
conditioned on policy so that's like
how things will actually turn out um
of this log part of the generative model
that is
the observations given the policy that's
how things will go
so maybe there's other interpretations
or other
parts that are really critical but i
think this is one of the key
formulations which is that there's a
policy related optimization problem
that can be phrased in a variational way
that is going to be bounded through a
free energy like
strategy to
estimate tractably some
other function q something related to

that
of basically what we would want to know
which is like
okay the observation that i look into my
bitcoin wallet and there's 15 bitcoin
how what policy do i have to take to
see that observation or for that counter
factual to exist
if only one knew so that's why there's
this tractable
boundable equation potentially i i i
think there's a lot to
say here and alec and anyone else i'd
appreciate any other
input but that was just one read on it
okay
so that was most of the part that i
wanted to go through at that level of
depth but then just to convey
the other sections and what they do for
the paper
but not go into every equation as much
and then talk about the figures
so 3.2 is the learning and inference
section
and in order to actually implement
this optimization scheme as it's laid
out
it turns out there has to be an update
process so
is it going to be a sort of uh stick
with what has worked or is it going to
be
novelty search those are kinds of things
that optimization
algorithms have to navigate the
trade-offs that are implicit uh
with different strategies depending on

different problems
and so this section defines how this
f uh variational free energy is defined
through time
and i'm not going to cover the
implementation details but alex
alec would be cool if you um you know
showed us a simulation or something like
that
uh 3.3 is about policy selection here's
what they write
under active inference policy selection
is achieved by updating q
of policy in order to minimize free
energy at fancy f
given the prior belief that policies
minimize expected free energy
i.e the p of policy is related to
this choice soft max related to the
um negative g negative uh
expected free energy of policy spaces
with short
as specified in equation three free
energy is minimized
when the uh q distribution of policy
is the same as the optimal free energy
minimizing uh expectation of policies
across policies like just like the free
energy optimal policy selection
for discrete action spaces with short
temporal horizons
basically g of policy can be evaluated
in full by considering each possible
policy so that's like connect four
you could just run out or tic-tac-toe
you can run out
every single option for the games at
that branch point

compute the absolute value of every
single possible outcome
and then make your choice however in
continuous action spaces there are
infinite policies meaning an alternative
approach is required
that's a really cool sentence because it
really conveys well
that actually the continuous action
space
even for a single variable is a truly
different domain
than a let's just say prisoner's dilemma
game
there's sometimes uh almost no overlap
in the algorithms that's why this paper
actually reflects quite in advance and
there's more details in that section
3.4 trajectory sampling so now how do
you go from just having this
distributional relationship between
policies and other uh variables like
related to generative models of the
world
and observation models how are they
going to
get there so to evaluate from the
distribution to the specific policy
so to evaluate the expected free energy
for any specific policy
first they have to evaluate the expected
future beliefs condition on that policy
so uh one can't really undertake
uh in this sort of normative framing
it's a secondary debate which we can
definitely have about to what extent
these are
experienced cognitively by humans but

we're just going to kind of
go with it with the intentional stance
and the sort of machine learning control
theory
take but and so beliefs it's not
necessarily cognitive
we're just thinking about what the agent
wants to do to succeed
but basically if the agent doesn't have
an expected future
belief sequence for that policy it will
be extremely difficult for them to
undertake it
and the trans um the fact that the
transition
model is probabilistic and the
parameters of the transition model or
random variables
induces a distribution over future
trajectories so
rather than just playing every possible
chess move
we're thinking about how we are fitting
a
more advanced kind of model well chess
is still discrete so
let's choose we'll see a specific
example
but this is just continuous optimization
later
in this figures but um this is like
having a distribution over future branch
points in a chess game
so whether it's discrete or continuous
you can still have this element of
uncertainty
of a distributional range over future
trajectories

in state spaces so several approaches
exist to approximate the propagation of
uncertain trajectories
for example one can ignore uncertainty
entirely and propagate the mean of the
distribution
or one can explicitly propagate the full
statistics of the distribution
so those are kind of two extreme
approaches one of them is just say yep
ensemble modeling is tracking the mean
and so i'm going to keep that moving
forward
and then the alternative is basically to
keep a summarization
of the entire distribution and use that
as what is being learned by the
population
in the current work it's utilized a
particle approach
whereby a monte carlo so sampling based
scheme
samples are propagated in particular we
consider
this big b samples from the parameter
distribution
data which is drawn from cube data
these sections convey how information on
policy selection
is accomplished in the model and how the
policy fitness landscape is explored in
order to
implement predictive control on
trajectories so the
policy through time implement it must
involve some element
of predictive control all right
then there's section um 3.5 3.6

so 3.5 has details on computing this
expected free energy
value as the title might suggest
in this section they describe how to
evaluate negative g
for a policy where they have used this
notational convenience to talk about
policies moving forward
so kind of just made it like a naked pie
rather than this upper superscript that
has to do with the time states um and
uncertainty
and the negative expected free energy
which is what we
are trying to get out here with the
negative g
is equal to the sum of this negative
expected free energies through
time and then that is where
a there's a decomposition that
not my area but just i would
think what it means is that there's a
decomposition
into this state information game
gain and an extrinsic value component so
this kind of breaks down into an explore
and exploit
components or some other trade-off we
see this a lot
with this negative g i would love it if
somebody could uh
talk a little bit why this is or if it
always is the case but
it always uh seems to be presented as
model accuracy penalized by model
complexity or
this plus this is it always two
parts in the equation is it ever just

one term can it be three terms
here it's three terms but what um
or four so i'm not exactly sure
why sometimes it has more or less terms
but that's what i'm curious about
then in 3.6 they write the model
presented
in the preceding sections serves as the
most general formulation applicable in
both partially observed and fully
observed environments
in what follows we describe an
implementation full of fur
implementation for the fully observed
case leaving an analysis of the
partially observed case for future work
so it's just another wrapper around the
fully observed case
to have the partial observations and
so they decided to use some benchmarks
that facilitated them
to basically have total observation and
this isn't like
cheating the examples they're gonna
describe
are kind of like balancing a dinner
plate on a stick
so in that case you do have total
observation of the model
so just because you can't control um uh
well
it may or may not be possible given a
certain stick and uh
motor reflex time but at the very least
you can observe everything so there's
also unobservable control theory
problems and then there's ones where the
system is actually

understandable and controllable like the
dinner plate but
you only get distorted or malicious
information
there's other ones where you get perfect
information but only at time points and
there's a game going on so
a lot of different ways that that can
play out but that's
sort of the uh the parts of the model
that can get switched out
and that's what also i think it will be
interesting to hear from any of the
authors on
so in section 4 they
go through some experiments and this is
kind of the
results phase of their paper and they
describe how they investigate
whether one whether the proposed active
inference model can successfully promote
exploration in the absence of reward
observations i.e exploration so that's
what's really difficult for the model
free
reinforcement learners to do is when
it's not getting the reward
it doesn't do very well can't explore
very well
because it's not winning and two whether
the model can achieve good performance
and high sample efficiency on
challenging continuous control tasks i.e
exploitation so that's what requires
tracking on something
we evaluate these two aspects of the
model separately leaving analysis of
their joint

performance i.e the exploration
exploitation dilemma
to future work so again show it
separately
in the kind of pure use case show that
it can do either one
and then it will be something like a
trade-off between the two of them or a
wrapper layer that will end up mediating
this trade-off
all right the explore task that they're
gonna do is called mountain car
and you're a little car on the bottom of
this valley
and you can push forward with
um the forward pedal or you can
push reverse with a reverse pedal and
you don't have a strong enough engine
to get up to the yellow flag with just
one
go it's like you're at the bottom of the
hill and you're in the
gear three seven on your bike but you
can go up a little bit and back a little
bit and so here you have to kind of
explore and figure out a strategy that
helps you explore more and more
in order to get enough speed by going up
this left hill in order to get
yellow flat and then the exploit tasks
that they're going to be talking about
are two-fold
there's the inverted pendulum task so
here we're still a little cart
or something like that and we can move
forward or backwards on a track
and then there's the pendulum that we're
trying to keep

upright and so that's like the dinner
plate on a stick
then there's the hopper task which is a
jumping
motor coordination task so let's look at
what they actually did for each of these
two
experiments and then see how the active
inference learner stacked up against
other algorithms so the mountain car
example is a one-dimensional track
positioned between two mountains the
goal is to drive up the mountain on the
right however the car's engine is not
strong enough to scale the mountain in a
single pass
similar to what set the only way to
succeed is to drive back and forth to
build up momentum so what was cool about
this
i thought was that the
code was uh on the openai site and on
their github.com
link is here and so it could be really
seen and played with and used as a
standard
so it seems pretty cool didn't know
about this resource
for standardizing different learning
algorithms
so how did the active inference agent
do well it did well in this test
otherwise they wouldn't have
made the paper and the way to read these
graphs
um a b and c are going to have the same
x and y axis
so i put the map

with the engine on the top
left so it kind of maps on to the bottom
left
panel a and so the position is related
to the x-axis
and so that's how far to the left and
how far to the right
that little train car goes and then the
velocity
is whether it's at rest zero whether
it's moving positively to the right or
whether it's moving negatively to the
left
so the reward agent who just sort of
pursues momentum
whichever way i guess um only ends up
getting a max speed of around 0.02 in
either way
and then only moves like .7 or something
in either direction the epsilon
greedy agent does manage to learn a
short range policy
that we can see brings it a little bit
further on one side
and it does take some rare excursions
into
slightly higher velocity regimes but in
the end it uh
doesn't really get too far like it
learns
how to go uphill when it's going uphill
but that only helps at a limited extent
whereas the active inference model as
implemented
has not just a more broad sampling
within the
um state-space conversions after 100
epochs

but their position
extends to much higher range in the x
and
also a higher velocity distribution
range in the y
so it was more able to get out of this
little ditch
by being an active inference learner
than by being
epsilon greedy or just sort of model
free reward based
all right the second optimization
example presented in the paper is this
hopper v2 again didn't know about this
resource
so this one is interesting it is
described as making a two-dimensional
one-legged robot
hop forward as fast as possible and the
code is also there i'm not going to put
the code up but
it kind of reminded me of this idea of
evolutionary computation
as control and so sometimes
there's so many angles to this first off
just this image
with a checkerboard um
floorboard it reminded me of some
simulators of evolutionary computation
like for linux and stuff like that where
you could have these blocky aliens that
would replicate and
take up a ton of computational resources
and just
clash into each other and sometimes it
was just the outline other times it was
3d blocks with this kind of a background
and a floor plan

but here and what we're evolving is
control strategies not
chromosomes representing genotypes or
let's just say
but actually a similarity is this idea
of the genetic algorithm and the genetic
algorithm
stochastic search because sometimes the
way that these control policies are
optimized
is actually through steps that involve
things like genetic search algorithms
that switch over large sets of their
parameters
so there's some just interesting
parallels and this i just thought was a
cool example
and at the intersection of motor
behavior
control but also complex state state
space
estimation even the beginnings of an
inactive
uh and maybe even embodied approach
because at some level
of movement complexity it's going to be
implicit in the system
and distributed in it just like other
systems that it has some notion of its
own affordances
if it had 500 muscles and it was a human
leg it would have some sort of uh
tensegrity structure that kept it
balanced in a way that facilitated
some types of actions but not others so
very interesting
choice i'd like to hear from
any authors or from other people like

what are other
benchmarks that are cool what other
control
systems problems are interesting or
applicable
what about social versions or networked
or communication based versions
so what were the results
okay so here is how to look at these
graphs
the x-axis is the epoch so that's just
the sampling through time that's how
many
generations of parameters are tried and
here the red lines
indicate the same interval of time so
c and d are related to epochs 1 through
100
and that's actually compressed up
against um a very small
uh unit on a and b and a
and c respectively are for this inverted
pendulum and um b
and d are for the hopper task so we can
talk about the same phenomena looking at
a and c in reference to the pendulum
and b and d in the hopper and in both of
these cases
zoomed in or zoomed out what is pretty
apparent
is that the act and the y-axis is reward
so that's just like
in the hopper task is how far you're
getting and then in the inverted
pendulum task
it's about um your time maintained given
a certain parameter combination for
policy and in both cases

the reward sharply increases
for the active inference agent and
also just immediately uh starts climbing
after some
sampling within the first 100 or even
the first uh
several epochs probably because of the
lower state space
of the uh cart because all the cart can
do
is move forward and backwards have a
policy on that like gas and break and
stuff like that
that's it's all cooked into the physics
of the pendulum
i wonder if there's a dump inverted
double pendulum task by the way that'd
be kind of interesting
and um in the hopper task there's
several more parameters at play
i didn't copy out the exact number but
there's several more so
it takes a couple more epochs reflected
by the zero
through um maybe 40 where the
the average is like not really getting
off the ground and this is being
contrasted against the ddpg which you
can read more about i'm just not going
to go into what this
algorithm is i just trusted all right
the authors used something that makes
sense
let's hear from anyone about what other
alternatives could be tested against
so those would all be things where i'm
just not as
up on literature but definitely would

welcome
the comments of someone from the machine
learning field and
what would be an impressive or useful
demonstration of active inference or
what would it mean
to do a control theory simulation
in this sort of a framework what else
was used
what what else is the current state of
the art for these inverted pendulum and
hopper
tasks just i i don't know
but it'd be great if anyone does know so
just to close out with the relationship
to the previous work
okay this is from their closing section
which maybe it's a
disciplinary thing but often
it's like a discussion section but if it
was relationship with previous work i
would have expected it to be a little
bit more contextualizing
and at the front of the paper so
but it read more like a discussion
related to
deep active inference which is like this
through timeness of active inference
they write our work builds upon these
previous models by incorporating model
uncertainty in its active resolution
we extend the previous point estimate
models to include full distributions
over parameters and update the expected
free energy functional
such that the uncertainty in these
distributions is actively minimized
this brings our implementation in line

with the canonical models of active
inference from the cognitive and
computational neuroscience literature
so this is almost like saying yeah one
more
moreover it enables us to evaluate the
feasibility of active exploration
under the scaled active inference
framework apply the model to more
complex control tasks
and obtain increased sample efficiency
relative to previous models
so it's kind of like saying we've made
this active inference model closer to
what it has been speculated to be
in the qualitative neurobehavioral realm
which i think is a good reason to be
reading it right now
because it does fit in nicely with our
discussions of behavior
and of um
neurobiology in the variational ecology
discussions so this is on the simulation
side how it comes to be this way
so this is kind of cool because deep
active inference is now being
implementable
through these kinds of models that are
presented in this paper
and previously was like oh yeah i guess
if it were deep active inference then it
would accommodate for narratives
and now this is the kind of framework
where oh yeah okay we could talk about
narrative through time
in this framework could we the second
area they address again another key word
for the paper

is model based reinforcement learning in
the current work we
opted for bayesian neural networks to
ensure consistency with the variational
principles espoused by the active
inference framework
but note that ensembles can be made
explicitly bayesian with minor
modifications
so they talked about several homologies
to reinforcement learning
and specifically amortized inference
there's actually one quote
that uh i pulled out from one of the
papers that are on the slide and that
talked about how
you might get the sense that this is
like a win-for-free situation with just
wrapping a functional
around the estimation of these
parameters because it's like oh it's a
neural network i'm implementing a neural
network so i have an additional degree
of freedom to learn non-linear policy
but then the paper argued that
actually you're constrained because the
functional is estimating
the mean and the variance on this
gaussian outcome
and so it doesn't actually liberate a
degree of uh of output
expressivity but rather
a definable and a scalable way to
implement
on current computational hardware a way
to implement this
massive pencil and paper problem of
estimating the gaussian mean and

variance of super super complex
high dimensional state spaces but that
still is what it's doing
so super interesting stuff and
um also the note here that the ensembles
could be explicitly bayesian so with
all the kinds of variants and flavors
that that introduces so
anyone who's a enthusiast or
expert in these areas i would appreciate
their perspective
on what that means or why this is
important or
interesting all right then info gain
and this was one of the key words too so
identifying scalable and efficient
exploration strategies remains one of
the key open questions in reinforcement
learning
model-free methods such as the greedy or
boltzmann choice rules
paging dr proton sean utilize
noise in the action selection process or
uncertainty in the reward
statistics a more powerful approach is
to construct a model of the world
allowing the agent to evaluate which
parts of the state space it has and has
not visited
this allows for measures such as the
amount of prediction error or prediction
or
improvement to be utilized for
exploration okay cool
if the learn model implicitly or
explicitly captures probabilistic
features
so that's the statistical regularities

of a niche then
information theoretic measures can be
used to guide exploration
so there's so much to say here and it's
very interesting
to bring in the information gain at the
end
really what it's saying is this learned
model
of active inference and the way that the
exploration is tied into the generative
model of the world such that action is
facilitated not just towards the
straightforwardly rewarding that's sort
of the model free rl
nor merely the um deep model
rewarding that's like i love running
marathons because it's healthy i'm not
you know saying it's not healthy just
it's it's the mindset that helps one
get there for sure and then um
this is saying we can actually go
another level beyond
that we can say with a deep generative
model
of what one is likely to be doing we can
maximize precision
which may be something you hear amongst
the you know ultra ultra
um committed of an area something like
this is just how and who i am so
in that sense it would be going beyond
this whole oh well this many
miles per day is healthy for me which is
a great motivation it might be
might be a multi-scale thing um just one
example
and definitely i know you know even for

me it's just changed like
through life so it just shows how it's
related to your niche your context your
social relationships
so many other areas and
just to kind of close on one thought on
this information again
um information gain surprised me at
first because i was like trying to
connect all this to information theory
and i realized that what it's about is
reducing uncertainty on target
parameters
information is about reducing your
uncertainty about things
and it's just about reducing our
uncertainty about something
else or about more specifically
something else being a multi-scale model
that includes some attributes that we're
probably pretty familiar with like the
observation model
some aspects that are homologous to
a reward model like sort of a preference
model let's just say but then some
aspects that are
mainly maybe new and that might be the
part about the free energy
and then that area as i understand it
is related to another type of
information theory and model selection
and the variational framework so if that
is an area
that anyone has insight into
i think it's pretty cool maybe we could
try to unpack that i'm not
sure what it looks like but uh
definitely

something i wondered about and minding
the p's and q's and
thanks alec for a little bit of the
email discourse about clarifying a few
aspects of the paper
because um it was pretty interesting
and i think there's a lot of
implications and what systems we can
possibly apply this to proximally or in
the medium term
so tons of interesting stuff we'll be
talking about this on november 10th 2020
and
the 17th at 7 30 am pst
and i think that is it and
let's see yeah one one closing thought
all we can have is a deep generative
model of the space intertwined with
action
so that the parameters we explore are
the ones that are at the trade-off
of successful and optimally informative
sometimes we fear
we veer towards successful which can be
not optimally informative
other times we are getting informed
greatly in a way that may not be
successful
but overall we manage that trade-off
well it's how we've gotten this far
so great work everyone thanks a lot for
listening
keep it up thank you for participating
we do provide follow-up forms to live
participants
and would be awesome if anyone had
feedback
or suggestions or questions

you can stay in communication with us
and up one more slide on that
parameters reparameterization trick and
the slides from that paper but
other than that thanks for listening
and for bearing with the uh anomalous
camera error but
yeah i'm looking forward to the end of
2020
coming in with all these cool
discussions we'll be having
we'd love to have some new participants
come online
or bring in some new perspectives that
we haven't considered
or talk about it from a beginner's
perspective
from any number of fields as a starting
point
just let us know that's the kind of
stuff we
just always appreciate hearing about so
have a good november and end of 2020
season and we look forward to
talking with you alright bye